HEART TWIN

A Personalized Digital Twin for Early Heart-Failure Deterioration Monitoring



Happiest Health Digital Twin Challenge 2026

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A synthetic-data monitoring prototype — not a clinically validated prediction system.

The problem — and our approach

Between clinical visits, a single measurement may not show how a person's condition is changing over time.

01 Historical context

Demographics and available diagnoses provide background context for monitoring.

02 Daily signals

Wearable-style HR, HRV, sleep, steps, SpO_2 and weight are simulated over time.

03 Personalized review

Compare recent readings with an individual baseline and explain key deviations.

HeartTwin's goal **A doctor-facing proof of concept for personalized monitoring and transparent risk signals.**

Architecture & data fusion

1	**Synthetic EHR**	One Synthea record + independently created synthetic profiles
2	**Wearable stream**	10 simulated patients • 30 days • 6 daily signals
3	**Personal baseline**	Compare recent observations against each patient's usual values
4	**Twin risk engine**	Rule-based score, trajectory, explanations and what-if scenario
5	**Streamlit dashboard**	Patient overview and clinician-oriented review panel

Data are synthetic/simulated; genetic markers are not included in the current implementation.

Personalized risk engine

The engine scores deviations across six wearable-style signals and adds context from recent risk trajectory.

Resting HR
Heart rate

HRV
Variability

Sleep
Duration

Steps
Activity

SpO_2
Oxygen saturation

Weight
Daily weight

Prototype-defined score categories

LOW	0–19
MODERATE	20–39
ELEVATED	40–59
HIGH	60–100

These are heuristic categories, not medically validated thresholds or probabilities.

ML experiments & model selection

Patient-level evaluation showed poor generalization, so the current prototype uses a transparent rule-based engine.

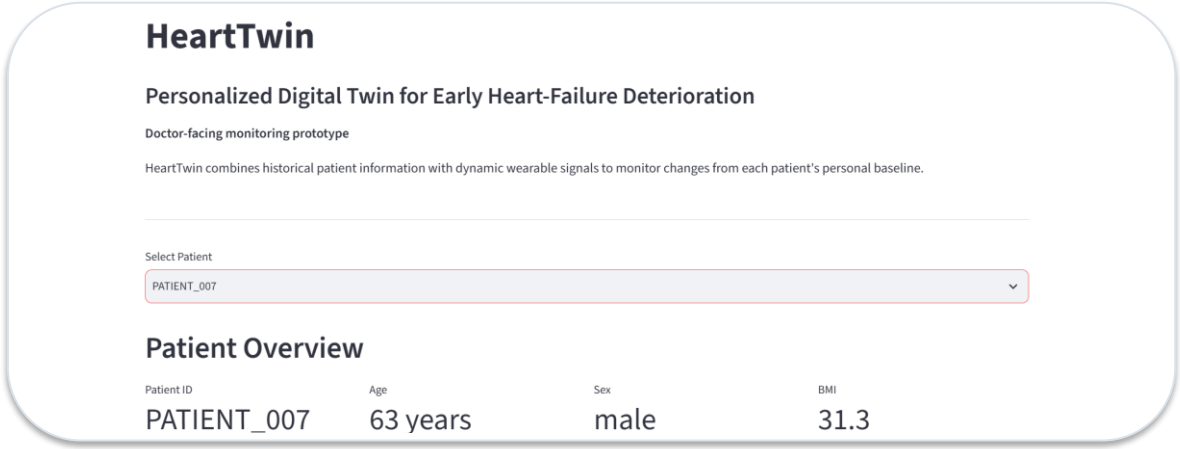
Experiment	Observed result	Interpretation
Logistic Regression	ROC-AUC 0.356 Positive-class F1 0.09	Weak performance on held-out patients
Random Forest	Positive-class recall 0.00	No positive cases detected at evaluated default threshold
GRU temporal model	ROC-AUC 0.347 Positive-class recall 0.00	Temporal model did not generalize in this dataset

Selected for this prototype **Interpretable Digital Twin Risk Engine (rule-based)**

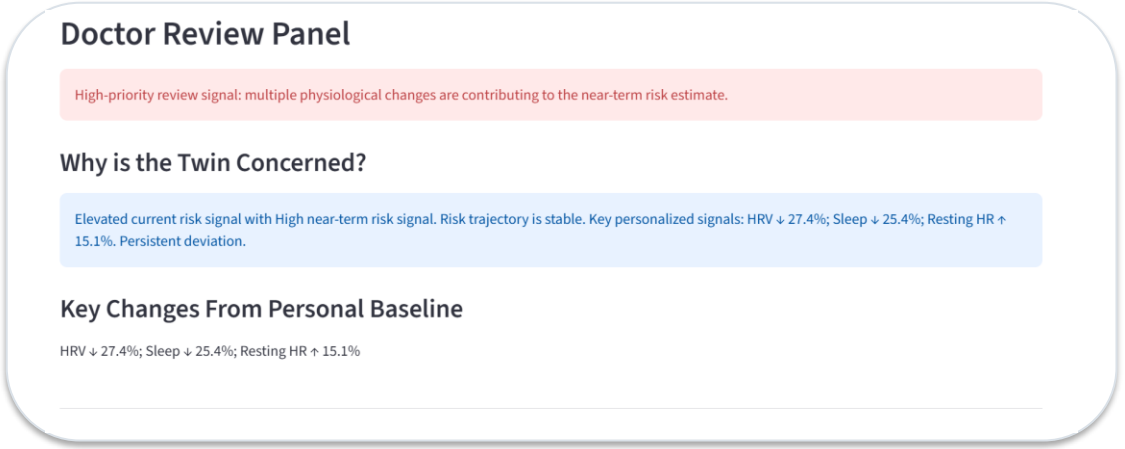
This is not a trained ML prediction model; future-event prediction remains an unmet development goal.

Working dashboard & what-if scenario

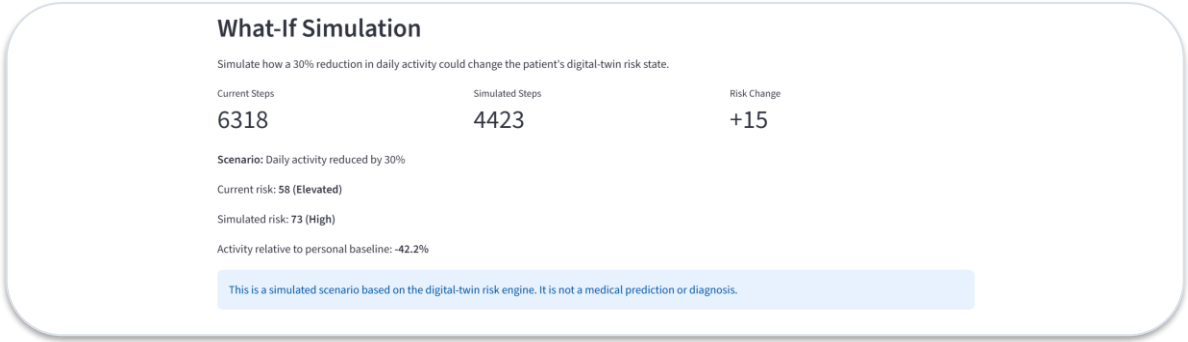
Actual Streamlit dashboard screenshots



Patient overview and current twin state



Doctor review panel and personalized deviations



What-if: 30% activity reduction

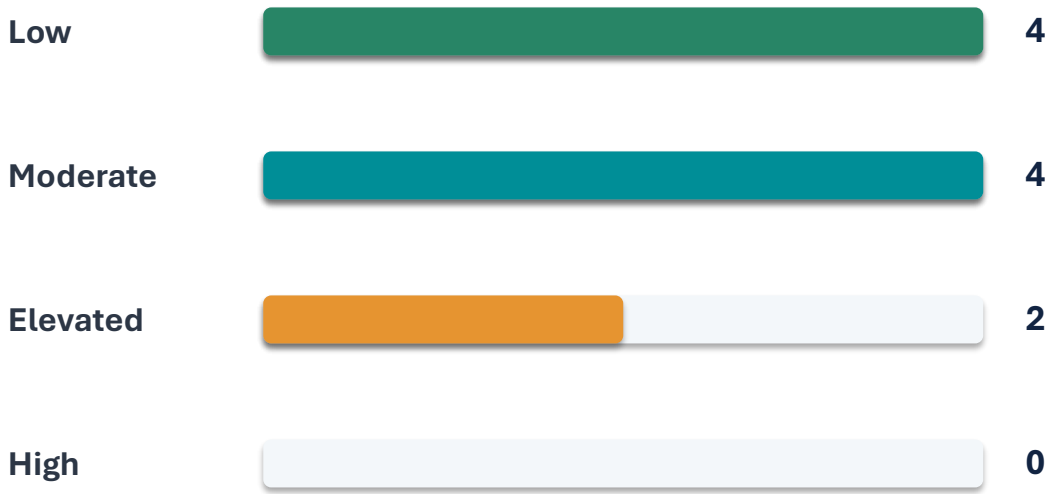
Example: PATIENT_007

Current score 58 (Elevated) → simulated score 73 (High); +15 points after reducing steps from 6,318 to 4,423.

Scenario output is generated by the prototype rules, not a medical prediction.

Validation — what the prototype shows

Risk-score distribution across 10 simulated patients



Mean risk: non-deteriorating

7.6

Mean risk: deteriorating

40.0

Elevated/High flags

0/5 non-deteriorating

2/5 deteriorating

What-if consistency check: risk increased in 10/10 scenarios; unchanged 0/10; decreased 0/10.

Small synthetic dataset; these are internal prototype checks, not clinical performance metrics.

From isolated measurements to personalized, explainable monitoring.

What is implemented

- Synthetic EHR context + simulated wearable time series
- Personal-baseline deviations and interpretable risk scoring
- Doctor-facing Streamlit dashboard and activity what-if scenario

Limitations to address

- Synthetic data and labels
- No genetic markers yet
- No clinical validation
- Rule-based score, not a trained event predictor

GitHub: github.com/KARIZMA-STEFFIROSE-E/HeartTwin_Pondicherry_University

Team details and 20+ minute demonstration-video link: add before submission.